

DANIYAL AHMAD

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SUMMARY

Computer Science student at Toronto Metropolitan University with hands-on experience building full-stack applications, AI-powered tools, and hardware-integrated systems. Strong foundation in data structures, software engineering principles, and system design, with additional experience in software business analysis and technical sales through work-integrated learning programs.

TECHNICAL SKILLS

Programming & Markup: Python, Java, JavaScript, C, C++, Bash, HTML, CSS

Academic Language Exposure: Lisp, Rust, Haskell, Smalltalk, Elixir

Web, UI & Systems: React, Node.js, REST APIs, Tailwind CSS, UX/UI Design Principles

AI, ML & Data: Retrieval-Augmented Generation (RAG), LLM APIs (Google Gemini), Vector Databases (ChromaDB), Sentence Transformers, Computer Vision, OpenCV, NumPy, Pandas, Matplotlib

Hardware & Embedded: Arduino Uno R3, Sensor/Actuator Integration (LEDs, Servo, Buzzer), Serial Communication

Tools & Platforms: Git/GitHub, Linux, VS Code, Vercel, Streamlit

PROJECTS

Codebase Q&A Assistant (RAG) | Python, ChromaDB, Sentence Transformers, Gemini API, Streamlit

- Built a Retrieval-Augmented Generation system that lets developers ask natural-language questions about an unfamiliar Python repository and get answers grounded in the actual source code.
- Used Python's ast module to parse repositories into structure-aware chunks (functions, classes, modules) with file/line metadata, embedded with Sentence Transformers and indexed in a persistent ChromaDB vector store.
- Implemented semantic retrieval and a grounded generation pipeline with Google Gemini that cites exact file and line sources and refuses to answer outside retrieved context, with retry/backoff for API errors.
- Built a Streamlit chat interface with source inspection and history; evaluated on **15 hand-written questions**, reaching **14 correct, 1 partial, and zero hallucinations**, then closed a retrieval gap found during evaluation.

AI-Based Face Recognition Attendance System | Python, OpenCV, face_recognition, Arduino

- Built a real-time facial recognition attendance system using Python, OpenCV, and the face_recognition library to detect and identify individuals from a live camera feed.
- Integrated an Arduino Uno R3 for physical feedback: green LED and a servo-controlled gate on a recognized face, red LED and buzzer alert on an unrecognized face.

Personal Portfolio Website | React, JavaScript, Tailwind CSS

- Designed and built a personal portfolio site with React and Tailwind CSS, deployed live on Vercel, with responsive, component-based layouts.

EXPERIENCE

Software Scope Development — Fraser Delivery (Riipen Level UP Project) | Riipen

May 2026

- Conducted a needs assessment of Fraser Delivery, a Canada–India courier company, analyzing its shipment booking, tracking, and manifest processes.
- Authored a software functionality outline and workflow diagram, then delivered a final scope of work document with requirements, a data model, user stories, and acceptance criteria. Certificate of completion awarded.

Sales & Business Development — Tahpin Inc. | Venture for Canada Intrapreneurship Program

July 2026 – Present

- Supporting Tahpin Inc.'s go-to-market initiative on search optimization for the tourism and retail space through outreach to local businesses.
- Building and maintaining a sales pipeline, including prospect research, outreach tracking, and documentation templates.
- Participating in a student team for Venture for Canada's Responsible AI Adoption Challenge, identifying a workplace process suited to a responsible AI solution.

CERTIFICATIONS

- CAE Immersion Program 2026 — Certificate (May 2026)
- Riipen Level UP Project Certificate — Fraser Delivery Software Scope Development (May 2026)

EDUCATION

BSc — Computer Science | Toronto Metropolitan University

September 2024 – Expected 2028

- Achievements: Dean's List (2024–2025), Dean's List (2025–2026), GPA: **3.5**